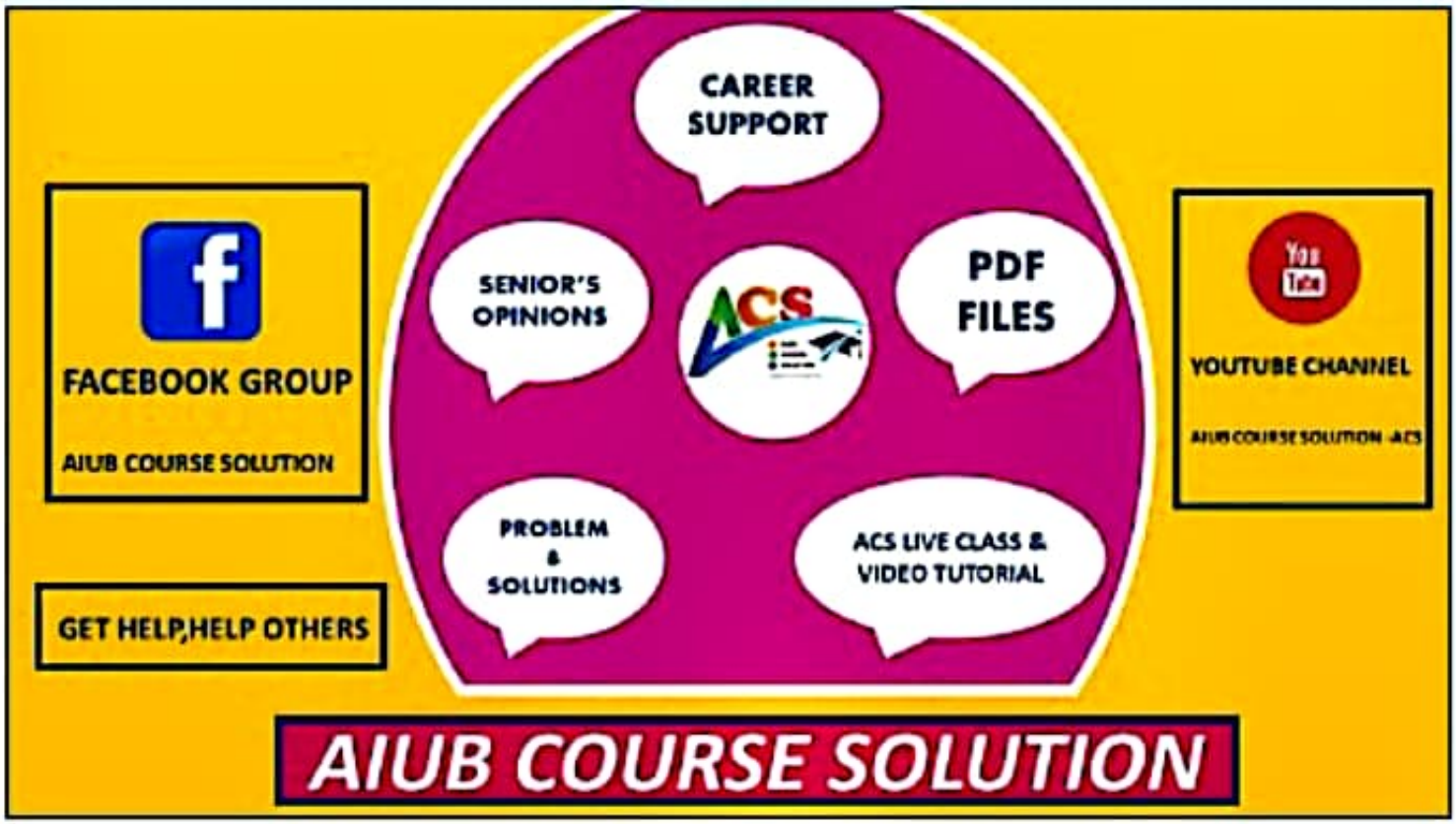


TOGETHER WE CAN ACHIEVE MORE
MOREMORE

COURSE NAME: OOAD

CHAPTER: lec 2

SOLVED BY

NAME: Zidul Alam Zim

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Lecture 2

UML

UML → Unified Modeling Language

OMG → Object Management Group

- OMG based on work from Booch, Rumbaugh, Jacobson.
- UML is OMG standard.

Use of UML → express and design documents, Software

- i) Particulary useful for object oriented (OO) design.

ii) UML is not a process.

iii) Independent of implementation language.

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* Standardized graphical notation for
  i) Specifying
  ii) Visualization
  iii) Constructing
  and iv) Documenting Software System.
* Increase understanding or communication of product to customers and developers.

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USE OF UML

For Visualizing ——

* best modeled textually ; others graphically.

* more than just a bunch of graphical symbols.

* One developer can write a model in the UML and another developer, or even another tool, can interpret that model unambiguously → (নির্দিষ্ট)

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For specifying —

* Specifying means → building models.

* UML addresses the Specification of all important analysis, design and implement decisions

  - Precise - নিখুঁত
  - Unambiguous → স্পষ্ট
  - complete - সম্পূর্ণ

↳ must be made in developing and deploying a software - intensive system.
(নিবিড়/তীব্র)

Deploying - বিস্তার করা Deploy - স্থাপন করা

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For constructing —

[UML is not a visual programming language]

UML → directly connected to a variety of programming languages.

Possible to map from a model in UML to Java / C+ or even table in RDBMS.

For documenting —

* UML address the documentation of a system's architecture and details

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* Provides a ~~langueg~~ language for expressing requirements and for tests.

* Finally, the UML provides a language for modelling the activities of project planning and release management

প্রকাশনা অভিজ্ঞান

Building Blocks of UML

Encompass → পরিবেষ্টন করা

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The vocabulary of the UML encompasses three kinds of building blocks

i) Things —— abstraction

ii) Relationships —— tie together

iii) Diagrams —— group interesting things

First class citizens in a model

→ abstraction.

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Things in UML

| → 4 types

Structural | Behavioral | Grouping | Annotation (Color)

Structural Things

* nouns of the UML.
* mostly statics parts of model.
* represent conceptual or physical elements
* Eight (8) kinds of — (i) class
  (ii) Interface

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Chain of responsibility (iii) Collaborzation

(iv) Use case

(v) Active class

(vi) component

↳ object

(vii) Artifact ⇒ নিদর্শন

(viii) Node → Vertex

Component: A physical and replaceble part of the system, that conforms to and provides the realization of a set of interfaces



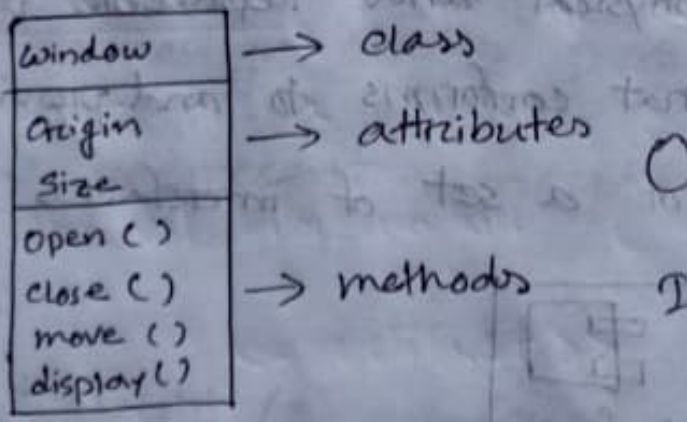
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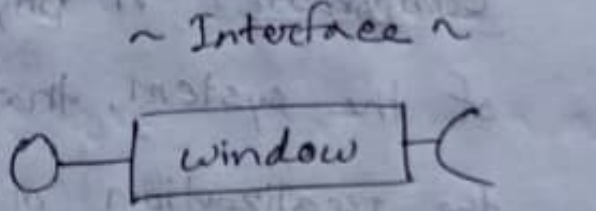
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Class: A description of set of objects, that share the same attributes, operations, relationships and semantics.

Interface: A collection of operation, that specify a service of a class or component.

* Interface describe the externally visible behaviour of that element.





I Window

I Paint

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Collaboration: It defines an Interaction and it is a society of roles and other elements, that provides some co-operative behaviors; that is bigger than the sum of all el elements → chain of responsibility

Use case: A description of set of sequence of actions that a system performs, that ye yields an observable result of value to a particular actor.

Active class: Such a class whose objects own one or more process or threads, that can initiate control activity.

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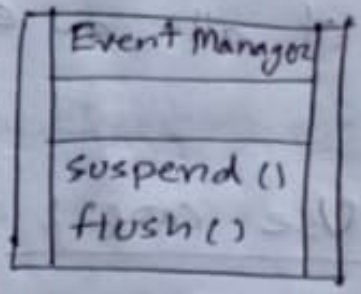
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* An active class is just like a class expeet that — its objects represent elements, whose behaviour is concurrent with other elements.

concurrent → happening at the same time



Artifact: A physical and replaceable part of a system (objeet) that contains physical information ("bits")

<<artifact>>
window. dll

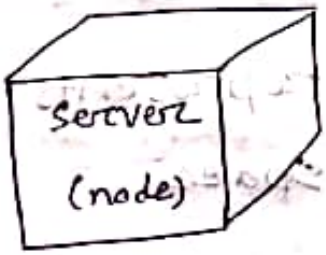
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Node: A physical element that exists at run time and represents a computational resource; generally having at least some memory and often processing capability.



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Behavioral Things

* dynamic parts of UML models
* Verbs of a model.
* represents behavior over time and space.

Operation

* Three kinds of — (i) Interaction
  (ii) A state Machine
  (iii) Activity

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Interaction : A behavior that comprises a set of messages exchanged among a set of objects within a particular context to accomplish a specific purpose.

comprise —— include ; context — ~~[illegible]~~

accomplish —— gain / conclude

display



* focus on the set of objects that interact.

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Grouping Things

* Organizational parts of UML models
* Compared with "box"
* Model ~~can~~ can be decomposed – ~~frameworks~~

Primary kind of grouping thing → packages

* kinds of packages – i) models

ii) frameworks iii) Sub systems.

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Frame - Work: A basic structure of a system

Package

* A general - Purpose mechanism for organizing elements into groups
* Purely conceptual; It only exists at development time.

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